

Riley Strode

VLSI Lab 1 DAC

Schematic:

- The DAC output resistance is equal to $2R \cdot 1R / (2R + 1R)$, which equals $2R/3$. Since $R=10k$, the output resistance of this DAC is $2 \cdot 10k/3 = \mathbf{6.67k}$
- The delay of this DAC = $0.7 \cdot R_{out} \cdot C_l = 0.7 \cdot 6.67k \cdot 10p = \mathbf{46.69 \text{ ns}}$
- Simulations: TBD (I need to figure out LTspice implementation with ElectricBinary)
- Driving Load: $R_l / (R_l + R_{out}) = 10k / (16.67k) = \mathbf{60\% \text{ of ideal output}}$

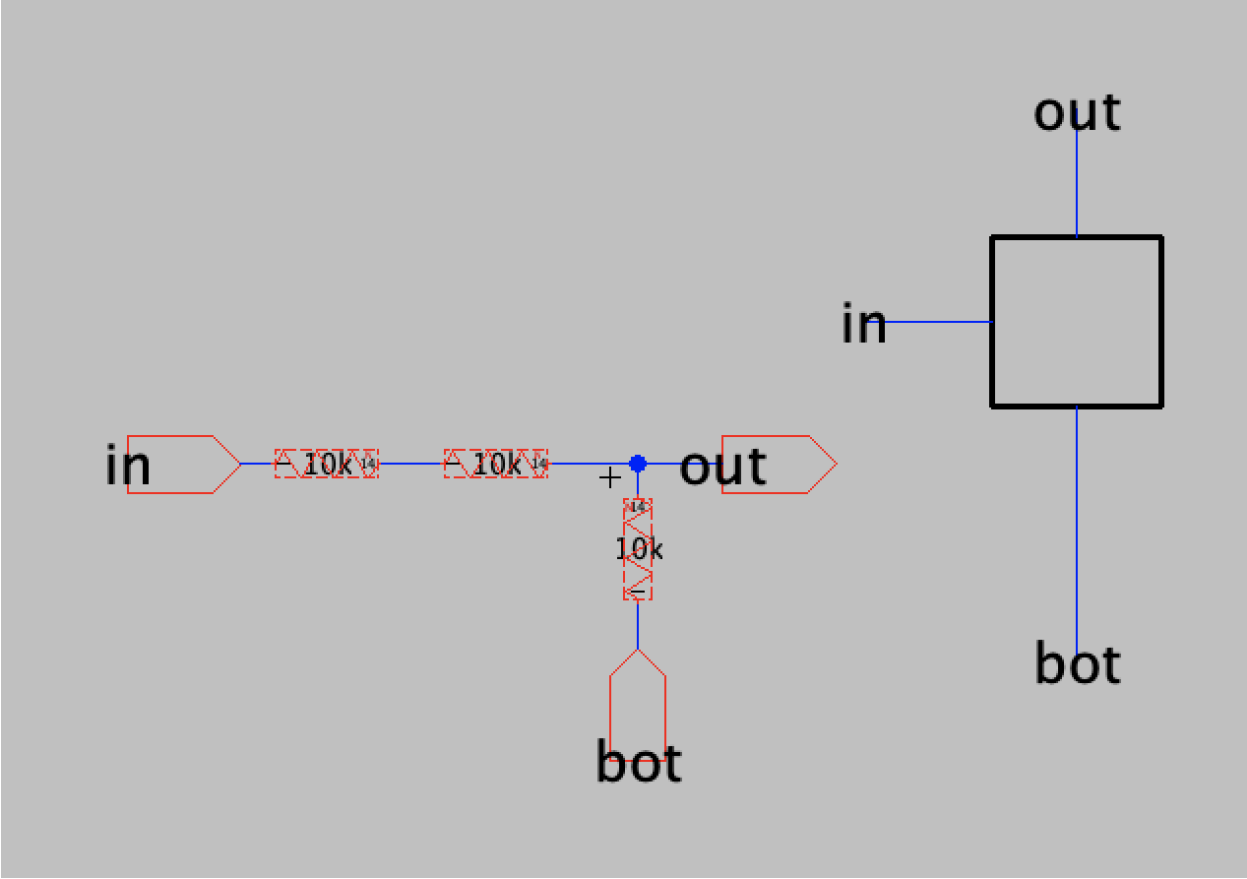


Figure 1: Single Voltage Divider

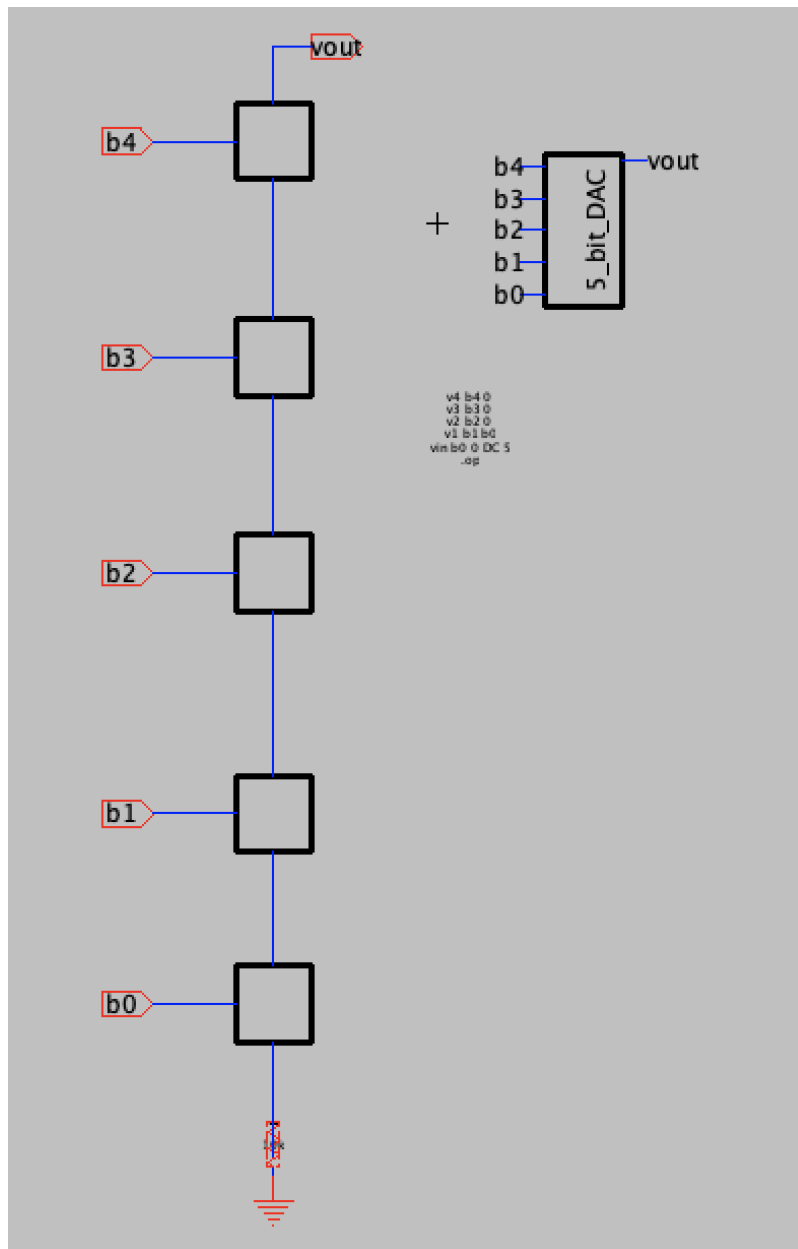


Figure 2: Cascaded Voltage Dividers to form DAC

Layout:

- Resistor Length and Width: Taking the width to be 14, the Rsquare = 855, and the desired output resistance to be 10k, $L = 10k \cdot 14 / 855 = \mathbf{164 = Length}$

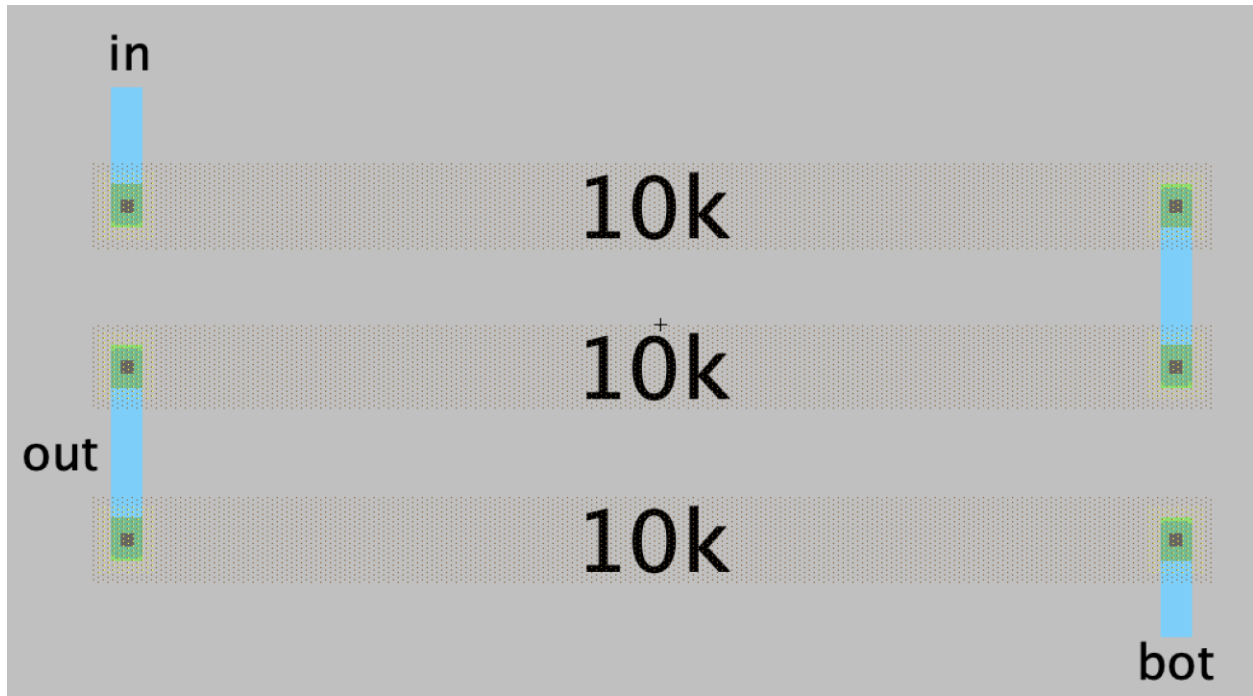


Figure 3: Voltage Divider Layout

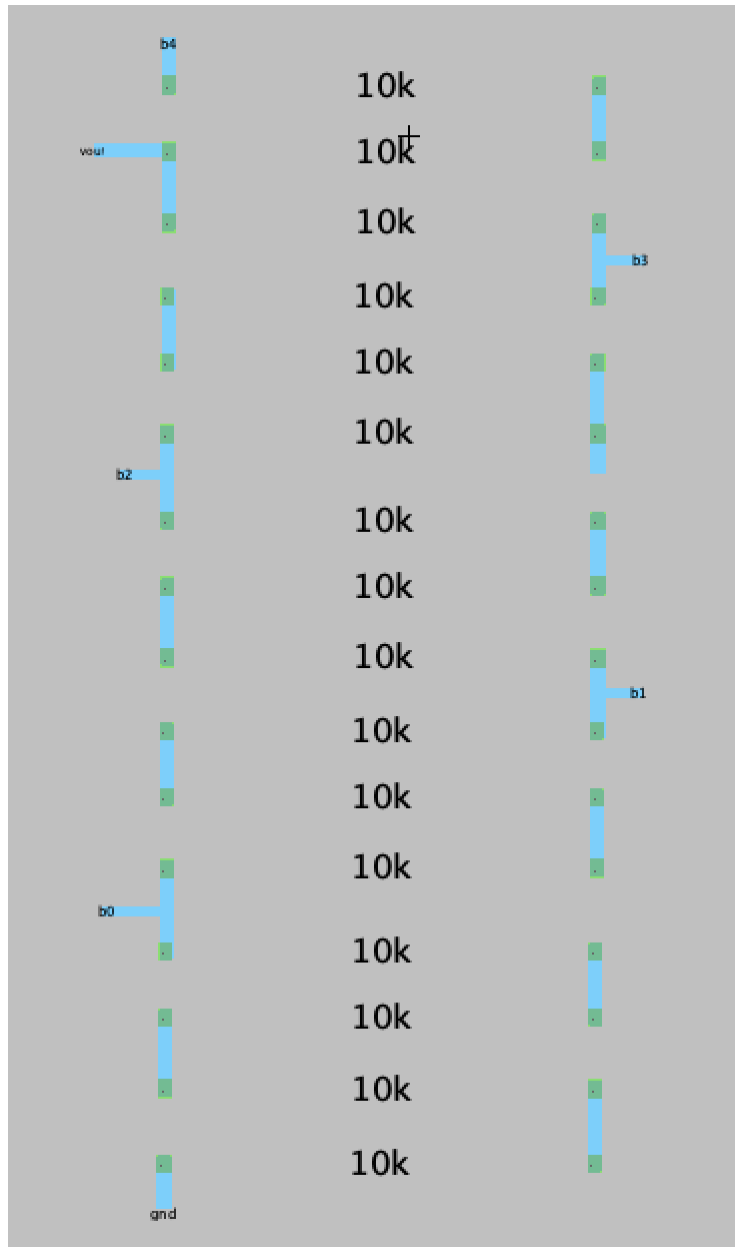


Figure 4: DAC Layout (Stacked Parallel)

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Running DRC with area bit on, extension bit on, Mosis bit  
Checking again hierarchy .... (0.0 secs)  
Found 18 networks  
Checking cell '5_bit_DAC{lay}'  
    No errors/warnings found  
0 errors and 0 warnings found (took 0.017 secs)
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Figure 5: Error Check